

1. The function f is plotted at Figure 1. This figure gives us the intuition that the function is convex.

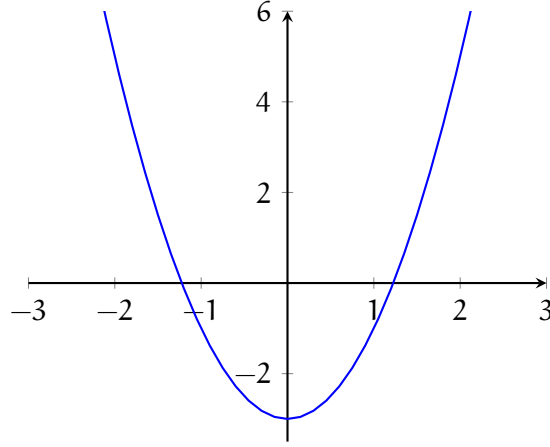


Figure 1: $f(x) = 2x^2 - 3$

We use the definition of convexity to formally show it. Let $x, y \in \mathbb{R}$ and $\lambda \in [0, 1]$. In order for $f(x)$ to be convex, it must satisfy the following condition:

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y).$$

We first write the inequation and use algebra to simplify it.

$$\begin{aligned} 2(\lambda x + (1 - \lambda)y)^2 - 3 &\leq \lambda(2x^2 - 3) + (1 - \lambda)(2y^2 - 3) \\ 2\lambda^2 x^2 + 4\lambda(1 - \lambda)xy + 2(1 - \lambda)^2 y^2 - 3 &\leq 2\lambda x^2 - 3\lambda + 2y^2 - 3 - 2\lambda y^2 + 3\lambda \\ 2\lambda^2 x^2 + 4\lambda(1 - \lambda)xy + 2(1 - \lambda)^2 y^2 &\leq 2\lambda x^2 + 2y^2 - 2\lambda y^2 \\ 2\lambda^2 x^2 + 4\lambda(1 - \lambda)xy + 2y^2 - 4\lambda y^2 + 2\lambda^2 y^2 &\leq 2\lambda x^2 + 2y^2 - 2\lambda y^2 \\ 2\lambda^2 x^2 + 4\lambda(1 - \lambda)xy - 4\lambda y^2 + 2\lambda^2 y^2 &\leq 2\lambda x^2 - 2\lambda y^2 \\ 2(\lambda^2 - \lambda)x^2 + 4\lambda(1 - \lambda)xy - 2\lambda y^2 + 2\lambda^2 y^2 &\leq 0 \\ 2(\lambda^2 - \lambda)(x^2 - 2xy + y^2) &\leq 0 \\ 2(\lambda^2 - \lambda)(x - y)^2 &\leq 0. \end{aligned}$$

Since $\lambda \in [0, 1]$, the quantity $(\lambda^2 - \lambda)$ is not positive. As $(x - y)^2 \geq 0$, for any $x, y \in \mathbb{R}$, the final inequation is always true, which makes the starting inequation true as well. With this, we conclude that $f(x)$ is a convex function.

2. The function is plotted at Figure 2. It gives us the intuition that the function is not convex, and not concave. We show it.

In order for $g(x)$ to be convex, it must satisfy the following condition:

$$g(\lambda x + (1 - \lambda)y) \leq \lambda g(x) + (1 - \lambda)g(y),$$

for each $x, y \in \mathbb{R}$ and each $\lambda \in [0, 1]$.

Similarly, in order for $g(x)$ to be concave, it must satisfy the following condition:

$$g(\lambda x + (1 - \lambda)y) \geq \lambda g(x) + (1 - \lambda)g(y),$$

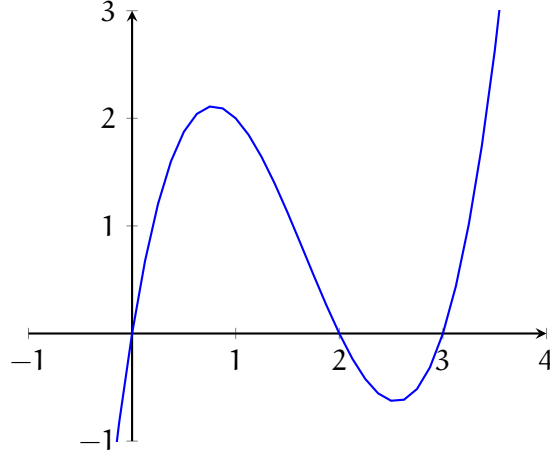


Figure 2: $g(x) = x^3 - 5x^2 + 6x$

for each $x, y \in \mathbb{R}$ and each $\lambda \in [0, 1]$.

To show that the function is not convex, and not concave, we need to prove the existence of x, y and λ that violate the above inequalities. Let us first rearrange the function in the following manner:

$$g(x) = x^3 - 5x^2 + 6x = x(x - 2)(x - 3).$$

From here, we see easily that $g(x) = 0$ for $x_1 = 0$, $x_2 = 2$, and $x_3 = 3$.

Let us further observe the function values in the points x_1 , x_2 , and $\lambda x_1 + (1 - \lambda)x_2$, where $\lambda = 0.5$ for the sake of simplifying the proof:

$$\begin{aligned} g(x_1) &= g(0) = 0 \\ g(x_2) &= g(2) = 0 \\ g(\lambda x_1 + (1 - \lambda)x_2) &= g(1) = 2 \end{aligned}$$

Since

$$g(\lambda x_1 + (1 - \lambda)x_2) > \lambda g(x_1) + (1 - \lambda)g(x_2),$$

we can conclude that $g(x)$ is not convex.

Let us further observe the function values in the points x_2 , x_3 , and $\lambda x_2 + (1 - \lambda)x_3$, where $\lambda = 0.5$, again for the sake of simplifying the proof:

$$\begin{aligned} g(x_2) &= g(2) = 0 \\ g(x_3) &= g(3) = 0 \\ g(\lambda x_2 + (1 - \lambda)x_3) &= g(2.5) = -0.625 \end{aligned}$$

Since

$$g(\lambda x_2 + (1 - \lambda)x_3) < \lambda g(x_2) + (1 - \lambda)g(x_3),$$

we can conclude that $g(x)$ is not concave either.